

Yiwei Gong

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SUMMARY

PhD candidate specializing in statistics and machine learning with expertise in probabilistic graphical models and variational inference. Experienced developing hierarchical and dynamic models with applications in biostatistics, healthcare, and financial data analysis. Currently focusing on Bayesian methods for time series and network embedding.

EDUCATION

PhD in Statistics and Data Sciences, The University of Texas at Austin, TX, **GPA 3.99** Aug 2020 – Dec 2025 (Expected)

Master of Statistical Science, Duke University, NC, **GPA 3.9** 2018 - 2020

BSc in Economics and Mathematics, University of Bristol, UK, **First Class Honours** 2015 - 2018

SKILLS

IT SKILLS: Python (PyTorch), R; **LANGUAGE SKILLS:** Chinese (**Native**), English (**Full Professional**), German (**Intermediate**)

AWARDS

- Kahmer Award 2025 (\$4500), UT Austin, in recognition of scholar merit, artistic ability and leadership aptitude.
- Best poster award, International Society of Bayesian Analysis World meeting (ISBA 2024)
- Young Researcher Travel Award, ISBA 2024 (\$800)
- Interdisciplinary Contest in Modelling 2017 Meritorious Winner (**international contest, Top 5%**).

PUBLICATIONS AND RESEARCH

Detecting State Changes in Functional Neuronal Connectivity using Factorial Switching Linear Dynamical Systems, Yiwei Gong, Susanna Mierau, Sinead Williamson (*Under Review*) <https://arxiv.org/abs/2411.04229>.

- **Presented** at WiML 2022 (NeurIPS affinity workshop), ISBA World Conference 2024 (poster and **oral**), Royal Statistical Society International Conference 2025 (selected as **contributed talk**).

Bayesian Geometric Network Embedding, Yiwei Gong, Catherine Calder (*working paper*)

Investigated hyperbolic continuous latent space models for networks, bridging the missing piece between hyperbolic random graphs and tree-like graphs; designing MCMC and VI methods for network embedding.

WORK EXPERIENCE

Computer Science Graduate Intern, Memorial Sloan Kettering Cancer Center, New York, NY May 2024 - Aug 2024

Developed a Gaussian process and VAE model to infer cell fate dynamics, incorporating causal intervention.

Oncology Bioinformatics Summer Intern, Genentech, San Francisco CA May 2023 - Aug 2023

- Proposed and executed a cutting-edge scalable Bayesian method for tumor mutation phasing by employing hierarchical modeling and variational inference.
- Achieved a computation speed 100-fold faster than the baseline model while maintaining robust model performance across both simulated and real-world datasets.
- Improved adaptability to intricate scenarios, leading to enhanced neoantigen targeting.

Data Scientist Research Intern, Duke Human Vaccine Institute, Durham, NC May 2019-March 2020

- Inferred the diverse immunogenetic characteristics of B cell repertoires in HIV-1 uninfected individuals; extracted B-Cell DNA fragments, subsequently identifying recombination gene segments and other stochastic components.
- Developed a generative model for the VDJ recombination process; constructed a hierarchical Bayesian network to estimate the occurrence probability of specific target sequences; novel discovery of the insertion region distribution.

Machine Learning Research Assistant, Bristol, UK May 2018 - Aug 2018

- Curated, cleaned, and summarized the large, sparse Yelp Dataset, provided degree distributions and local graphs, utilized sparse SVD and t-SNE for high-dimension embedding, clustering and data visualization.

Teaching Assistant for statistics, machine learning and optimization, multiple locations Jan 2018 - Present

PROJECTS, TEAMWORK AND LEADERSHIP EXPERIENCE

Graduate Research Assistant, Austin, TX

Cleaned and analyzed >500K entries of university-wise asymptomatic covid test data. Formulated a Gaussian mixture model for clustering test results and imputed highly censored data using the MCMC and EM algorithms.

Stock Index Data Analysis during Pandemics, Durham, NC

Applied time-varying vector AR processes to analyze major stock indices in three pandemic periods, practiced FFBS for posterior parameter inference with forecasting; latent factor PCA to evaluate the stochastic volatility.

SDS Computing committee member, Austin, TX